INTELLIGENT AGENTS: GAME THEORY WORKSHEET (II)

1. Question 1

Find a game that has at least one equilibrium, but in which iterative elimination of dominated strategies yields a game with no equilibria.

2. Question 2

Consider the following game:

		Player 2	
		L	R
Player 1	T	3, 1	0, 4
	B	2, 3	1, 1

(1) Find weakly and strictly dominated strategies, if any.

(2) Compute pure strategy Nash equilibria.

(3) Suppose Player 1 has a mixed strategy σ_1 such that $\sigma_1(T) = \frac{1}{3}$ and Player 2 has a mixed strategy σ_2 such that $\sigma_2(L) = \frac{3}{4}$: compute the expected utility of both players given the above mixed strategies and check whether those strategies form a mixed Nash Equilibrium.

(4) Find mixed Nash equilibria using the best response method.

3. Question 3

Consider the following game:

		Player 2	
		L	R
Player 1	T	5, 2	3, 1
	B	4, 1	3, 3

(1) Find weakly and strictly dominated strategies, if any.

(2) Compute pure strategy Nash equilibria.

(3) Compute all mixed Nash equilibria using the best response method.

4. Question 4

Consider the following game. Perform iterated elimination of strictly dominated strategies. What is the set of mixed equilibria of the game you obtain after the elimination process?

		Player 2				
		G	H	I	J	K
Player 1	A	7, 7	1, 6	3, 8	3, 10	5, 3
	B	11, 7	6, 7	1, 9	3, 10	7, 9
	C	1, 4	1, 4	1, 5	6, 3	4, 2
	D	11, 1	5, 2	8, 3	5, 4	8, 2
	E	7, 10	6, 6	7, 11	3, 10	5, 9
	F	10, 6	2, 6	7, 7	4, 8	6, 7